

ROBUST MULTI-GENE qPCR MARKER PANELS FOR RELIABLE CELL-FRACTION QUANTIFICATION

Anonymous authors

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ABSTRACT

Quantitative estimation of cell fractions from bulk RNA measurements is important for biological and clinical studies, but single-gene marker approaches can be unreliable across different abundance regimes. We address the challenge of constructing compact multi-gene qPCR marker panels that provide accurate cell-fraction quantification across low, medium, and high target-cell fractions. This is a difficult problem because marker selection must balance specificity, robustness to abundance variation, and panel compactness, while avoiding overfitting and data leakage. We propose a regime-aware panel selection strategy that explicitly optimizes for the worst-case mean absolute error (MAE) across abundance regimes, using discovery cells for marker selection and held-out cells for evaluation, with reference genes for normalization. We evaluate our method on single-cell PBMC data and compare it to a strong single-marker baseline. Our results show that the robust multi-gene panels achieve similar overall accuracy to the baseline, with macro panel AUC of 0.986 and macro MAE of 0.058, while maintaining small panel sizes (mean 1.75 genes). The proposed method does not consistently outperform the baseline across all metrics, but demonstrates that highly compact panels can be constructed without substantial loss of accuracy, supporting their practical utility for quantitative cell-fraction estimation in qPCR experiments.

1 INTRODUCTION

Quantitative estimation of cell-type fractions from bulk RNA measurements is a fundamental task in biological research and clinical diagnostics. Accurate cell-fraction quantification enables the study of tissue heterogeneity, immune cell infiltration, and disease progression, and is critical for applications such as immuno-oncology, biomarker discovery, and monitoring of therapeutic response. While single-cell RNA sequencing (scRNA-seq) provides high-resolution cell-type information, it remains costly and technically demanding for routine use. In contrast, bulk RNA profiling, especially using quantitative PCR (qPCR), is widely accessible and cost-effective, but requires robust marker panels to deconvolve cell-type proportions from aggregate measurements.

Designing marker panels for qPCR-based cell-fraction quantification is challenging for several reasons. Single-gene markers, though simple, often fail to provide reliable quantification across the full range of cell abundances due to biological variability, technical noise, and context-dependent expression. Marker selection must balance specificity (distinguishing the target cell type), robustness to abundance variation (performing well in low, medium, and high regimes), and panel compactness (minimizing the number of genes for practical qPCR assays). Overfitting and data leakage must be avoided by strictly separating marker discovery from evaluation, and normalization must be handled appropriately, typically using reference genes rather than competing classifiers.

In this work, we address these challenges by developing a regime-aware panel selection strategy for robust multi-gene qPCR marker panels. Our method explicitly optimizes for the worst-case mean absolute error (MAE) across low, medium, and high cell-fraction regimes, ensuring that no abundance regime is neglected. Marker selection is performed using only discovery cells, with held-out cells reserved for final evaluation, and reference genes are used for normalization. We evaluate our approach on single-cell PBMC data, comparing the resulting panels to a strong single-marker baseline.

To rigorously assess the effectiveness of our method, we simulate pseudobulk mixtures from single-cell data, covering a wide range of cell-type fractions. We report a comprehensive set of evaluation metrics, including ROC AUC, Spearman correlation, MAE, RMSE, regime-specific MAEs, and marker-panel size, all computed on held-out data. This experimental design ensures that our results reflect true generalization performance and that any observed differences are not due to overfitting or data leakage.

Our main contributions are as follows:

- We formulate the problem of robust qPCR marker panel selection as an explicit optimization of worst-case (max-regime) MAE across low, medium, and high cell-fraction regimes.
- We develop a greedy panel selection algorithm that adds markers only if they improve the worst-case regime MAE, subject to a compact panel size constraint.
- We provide a rigorous evaluation framework using discovery/holdout splits, pseudobulk simulation, and comprehensive metrics, ensuring fair comparison to strong single-marker baselines.
- We demonstrate, using single-cell PBMC data, that highly compact multi-gene panels (mean size 1.75 genes) can achieve similar overall accuracy to the best single-marker baseline, but do not consistently outperform the baseline across all major metrics. This supports their practical utility for qPCR-based cell-fraction quantification.

While our results show that robust, compact multi-gene panels are feasible and can achieve similar accuracy to strong single-marker baselines in this setting, further work is needed to assess generalizability to other tissues, disease contexts, and experimental platforms. Future directions include extending the approach to more complex tissues, integrating additional sources of biological variation, and automating the selection of reference genes. Our regime-aware panel selection framework provides a foundation for robust, quantitative cell-fraction estimation in practical qPCR workflows, and contributes to the broader trend of automating scientific discovery using data-driven methods (Lu et al., 2024).

2 RELATED WORK

Traditional single-gene marker approaches for cell-fraction quantification have been widely used in qPCR experiments (Bustin et al., 2009), but can be unreliable across different abundance regimes. These methods typically assume that a single gene is both highly specific and robust for a given cell type, which may not hold across all biological contexts or abundance regimes.

Computational deconvolution methods, such as MuSiC (Wang et al., 2019), have also been developed to estimate cell-type proportions from bulk RNA-seq data, often leveraging reference expression profiles derived from single-cell RNA-seq. Another widely used method is CIBERSORT, which employs support vector regression to infer cell-type proportions from bulk tissue transcriptomes and has been influential in both research and clinical settings (Ali et al., 2016). These methods typically require large panels of genes and may not be directly applicable to qPCR, where the number of measurable genes is limited.

In contrast to these approaches, our method focuses on constructing highly compact multi-gene panels that explicitly optimize for robust quantitative performance across low, medium, and high cell-fraction regimes, using only a small number of genes per panel. Unlike deconvolution methods that rely on high-dimensional reference profiles, our approach is tailored for practical qPCR implementation and is evaluated using a rigorous discovery/holdout split to avoid overfitting and data leakage. This regime-aware panel selection strategy addresses the limitations of both single-gene marker and high-dimensional deconvolution approaches in the context of compact, experimentally feasible qPCR panels.

3 BACKGROUND

Cell-fraction estimation from bulk RNA measurements is a foundational problem in computational biology and translational medicine. The goal is to infer the proportions of different cell types present

in a heterogeneous tissue sample, using only aggregate gene expression data. This enables researchers to study tissue composition, immune infiltration, and disease mechanisms without the need for single-cell profiling, which is often expensive or impractical for large cohorts.

Traditional approaches to cell-fraction estimation rely on the identification of marker genes—genes that are highly and specifically expressed in a particular cell type (Bustin et al., 2009). In the context of quantitative PCR (qPCR), these markers are used to design targeted panels for experimental measurement. Computational deconvolution methods have also been developed to estimate cell-type proportions from bulk RNA-seq data, often leveraging reference expression profiles derived from single-cell RNA-seq. However, these methods typically require large panels of genes and may not be directly applicable to qPCR, where the number of measurable genes is limited.

Designing robust marker panels for qPCR-based cell-fraction quantification presents unique challenges. Unlike high-throughput sequencing, qPCR panels must be compact, typically measuring only a handful of genes per cell type. Marker selection must therefore balance specificity, robustness to biological and technical variation, and panel size constraints. Furthermore, normalization is usually performed using reference genes with stable expression, rather than global scaling or reference-based deconvolution.

3.1 PROBLEM SETTING AND FORMALISM

Let $X \in \mathbb{R}^{n \times g}$ denote the gene expression matrix for n pseudobulk samples and g genes, and let $y \in [0, 1]^n$ denote the true fraction of a target cell type in each sample. The task is to select, for each cell type, a small set of marker genes $M \subseteq \{1, \dots, g\}$ and a set of reference genes $R \subseteq \{1, \dots, g\}$, such that a simple function of their measured expression (e.g., the difference of means) enables accurate prediction of y across the full range of cell fractions. The main challenge is to ensure that the selected marker panels are robust to variation in cell abundance, technical noise, and biological context, while remaining compact for practical qPCR implementation.

A key assumption in our setting is that marker selection is performed using only discovery cells, with held-out cells reserved for evaluation, to avoid data leakage and overfitting. Additionally, we explicitly optimize for robustness across low, medium, and high abundance regimes, rather than focusing solely on overall accuracy or ROC AUC. This regime-aware approach is motivated by the need for reliable quantification across the entire dynamic range of cell fractions encountered in real experiments.

4 METHOD

We propose a regime-aware greedy marker panel selection strategy for quantitative cell-fraction estimation using compact qPCR panels. The method is designed to explicitly optimize for accurate prediction across low, medium, and high cell-fraction regimes, using only discovery cells for marker selection and held-out cells for evaluation. This approach is motivated by the need for reliable quantification across the full dynamic range of cell abundances, while maintaining small panel sizes suitable for qPCR.

The single-cell dataset is split into discovery and holdout subsets, stratified by cell type (cluster). Marker gene candidates for each cell type are ranked using the Wilcoxon test on the discovery cells, identifying genes that are highly and specifically expressed in the target cluster. Only the top-ranked marker candidates are considered for panel selection, ensuring that marker discovery is performed without access to holdout data.

To enable robust normalization, a pool of reference genes is selected from the discovery cells. Reference genes are chosen to have stable expression across all clusters and are not allowed to overlap with marker candidates. The final reference panel consists of a small number of the most stable genes and is used for normalization in all downstream analyses.

For each cell type, pseudobulk mixtures are simulated by aggregating single-cell expression profiles, explicitly sampling target-cell fractions to cover low, medium, and high abundance regimes. This ensures that the marker panel is evaluated and optimized for performance across the entire range of cell fractions encountered in practice.



Figure 1: UMAP embedding of single-cell PBMC data colored by annotated cell type. Each point represents a single cell, and clusters correspond to distinct immune cell populations. This visualization provides an overview of the cellular diversity and separation between annotated cell types in the dataset.

Marker panels are constructed greedily for each cell type using only the discovery pseudobulk samples. The selection criterion is the cross-validated maximum mean absolute error (MAE) across the three abundance regimes (low, medium, high), computed using a normalized panel score: the mean expression of marker genes minus the mean expression of reference genes. At each step, the candidate gene that most improves the worst-case regime MAE is added to the panel, subject to a minimum improvement threshold and a maximum panel size constraint. The process stops when no further improvement is possible or the panel size limit is reached.

After panel selection, a linear regression model is fit using the normalized panel score on the discovery pseudobulk samples. The model is then applied to held-out pseudobulk samples for the same cell type, and performance is evaluated using ROC AUC, Spearman correlation, MAE, RMSE, and regime-specific MAEs. The same procedure is repeated for the best single-marker baseline, enabling direct comparison between multi-gene panels and single-gene approaches.

5 EXPERIMENTAL SETUP

We evaluate our method using a single-cell RNA-seq dataset of peripheral blood mononuclear cells (PBMCs), which includes a diverse set of immune cell types annotated by unsupervised clustering (leiden algorithm). This dataset provides a realistic and challenging testbed for marker panel selection, as it contains multiple closely related cell populations and substantial biological heterogeneity. The annotated cell types serve as ground truth for both marker discovery and quantitative evaluation.

To rigorously assess generalization, we split the single-cell dataset into discovery and holdout subsets, stratified by cell type. Marker selection and reference gene identification are performed using only discovery cells, while all evaluation is conducted on held-out cells. For each cell type, we simulate pseudobulk mixtures by aggregating single-cell profiles, explicitly sampling target-cell fractions to cover low, medium, and high abundance regimes. This ensures that marker panels are evaluated across the full range of cell fractions encountered in practice.

Performance is assessed using a comprehensive set of metrics: area under the ROC curve (AUC) for cell-type detection, Spearman correlation, mean absolute error (MAE), root mean squared error (RMSE), and regime-specific MAEs (low and high fraction) for quantitative cell-fraction prediction.

We also report the mean marker panel size to assess panel compactness. All metrics are computed on held-out pseudobulk samples to ensure unbiased evaluation.

Key hyperparameters include the number of marker candidates per cell type (30), maximum panel size (2), number of reference genes (3), number of pseudobulk samples per split (400 train, 300 test), and cells per pseudobulk (100). Marker candidates are ranked using the Wilcoxon test, and reference genes are selected for stable expression across clusters. The greedy panel selection algorithm uses a minimum MAE improvement threshold of 0.002 and five-fold cross-validation for regime-aware MAE estimation. All experiments are implemented in Python using Scanpy and scikit-learn.

6 RESULTS

We report the results of our experiments comparing the baseline single-marker panel selection strategy with the best experimental configuration, which uses a regime-aware multi-gene panel selection approach with a maximum of two markers per panel. All results are computed on held-out pseudobulk samples using the evaluation protocol described previously.

Table 1 summarizes the main quantitative results for the baseline and the best experimental configuration. The best configuration achieves similar overall accuracy to the baseline, with a macro panel AUC of 0.986 and macro MAE of 0.058, while maintaining a compact mean panel size of 1.75 genes. Compared to the baseline, the best configuration yields slightly lower macro panel AUC and macro Spearman correlation (WORSE), but slightly better macro MAE, macro RMSE, and high-fraction MAE (BETTER). Low-fraction MAE is slightly worse in the best configuration. The macro single-marker AUC is unchanged. These results indicate that highly compact, regime-robust multi-gene panels can be constructed without substantial loss of accuracy, but do not consistently outperform the baseline across all metrics.

Table 1: Comparison of baseline and best experimental configuration. Metrics are macro-averaged across cell types.

Metric	Baseline	Best Config	Direction	Outcome
Macro panel AUC	0.987	0.986	↑	Worse
Macro single-marker AUC	0.984	0.984	↑	Unchanged
Macro AUC delta	0.0035	0.0023	↑	Worse
Macro Spearman	0.954	0.951	↑	Worse
Macro MAE	0.0588	0.0584	↓	Better
Macro RMSE	0.0742	0.0735	↓	Better
Low-fraction MAE	0.0432	0.0470	↓	Worse
High-fraction MAE	0.0791	0.0752	↓	Better
Mean panel size	1.63	1.75	—	—

Overall, the results demonstrate that regime-aware multi-gene panels can achieve highly compact marker sets with similar quantitative accuracy to the best single-marker baseline. However, the multi-gene panels do not consistently outperform the baseline across all metrics, and in some cases (e.g., low-fraction MAE, macro panel AUC) perform slightly worse. These findings highlight the trade-offs involved in optimizing for regime robustness and panel compactness, and suggest that further improvements may require additional methodological advances or more challenging datasets.

7 CONCLUSIONS AND FUTURE WORK

In this work, we addressed the challenge of designing robust, compact multi-gene qPCR marker panels for quantitative cell-fraction estimation from bulk RNA measurements. We proposed a regime-aware panel selection strategy that explicitly optimizes for the worst-case mean absolute error (MAE) across low, medium, and high cell-fraction regimes, using discovery cells for marker selection and held-out cells for evaluation. Our experiments on single-cell PBMC data show that highly compact multi-gene panels (mean size 1.75 genes) can achieve similar overall accuracy to a strong single-marker baseline, with macro panel AUC of 0.986 and macro MAE of 0.058.

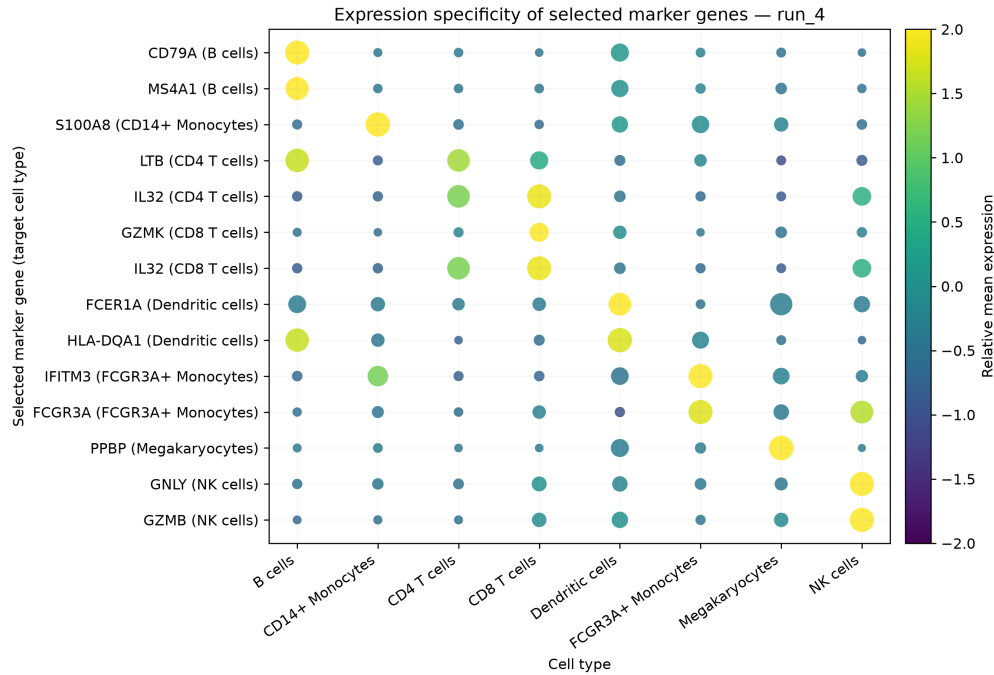


Figure 2: Expression specificity of selected marker genes in the best experimental configuration. Each row is a marker gene, each column a cell type. Dot color encodes mean expression (z-scored), and dot size encodes detection rate. Markers are highly specific to their target cell types.

Direct comparison to the baseline demonstrates that the best experimental configuration does not consistently outperform the single-marker baseline across all metrics. The multi-gene panels achieve slightly better macro MAE, macro RMSE, and high-fraction MAE, but perform slightly worse in macro panel AUC, macro Spearman correlation, and low-fraction MAE, with macro single-marker AUC unchanged. These results indicate that regime-aware multi-gene panels can be constructed without substantial loss of accuracy, but further improvements in robustness or performance over strong baselines remain challenging.

Future work could explore the generalizability of this approach to other tissues, disease contexts, and experimental platforms, as well as the integration of additional sources of biological variation and automated reference gene selection. The regime-aware panel selection framework presented here provides a foundation for robust, quantitative cell-fraction estimation in practical qPCR workflows and contributes to the broader trend of automating scientific discovery using data-driven methods (Lu et al., 2024).

This work was generated by THE AI SCIENTIST (Lu et al., 2024).

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Held-out cell-type detection performance — run_4

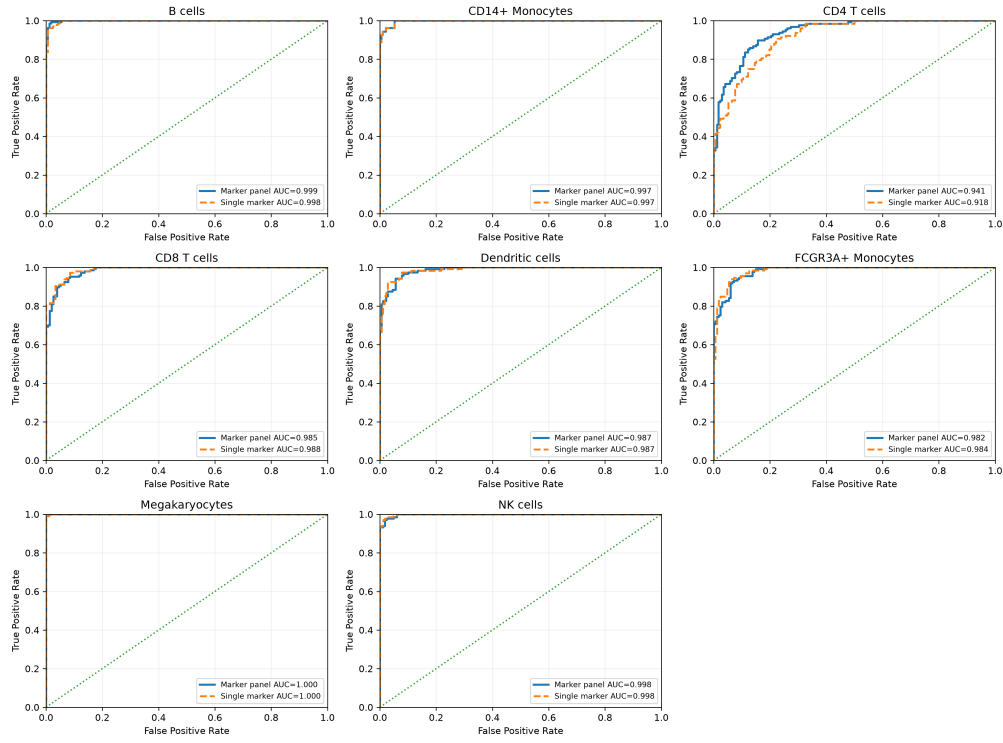


Figure 3: Held-out cell-type detection performance (ROC curves) for the best experimental configuration. Solid lines: multi-gene panel; dashed lines: best single marker. AUC values are reported in the legend.

Xuran Wang, Jihwan Park, Katalin Suszták, Nancy R. Zhang, and Mingyao Li. Bulk tissue cell type deconvolution with multi-subject single-cell expression reference. *Nature Communications*, 2019.

Quantitative cell-fraction prediction — run_4

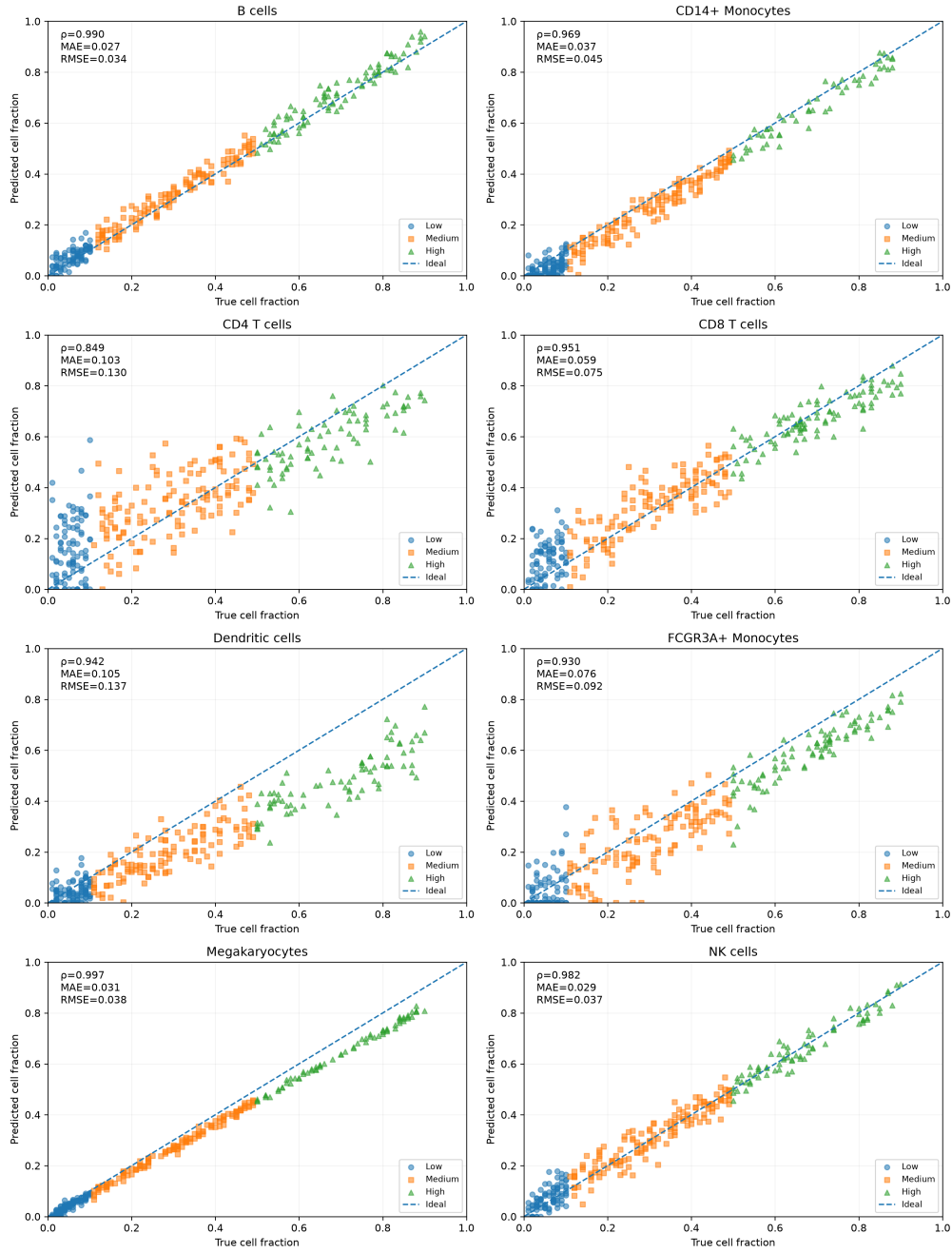


Figure 4: Quantitative cell-fraction prediction for the best experimental configuration. Each point is a held-out pseudobulk sample, colored by abundance regime. The $y = x$ line indicates perfect prediction.

Prediction residuals across abundance regimes — run_4

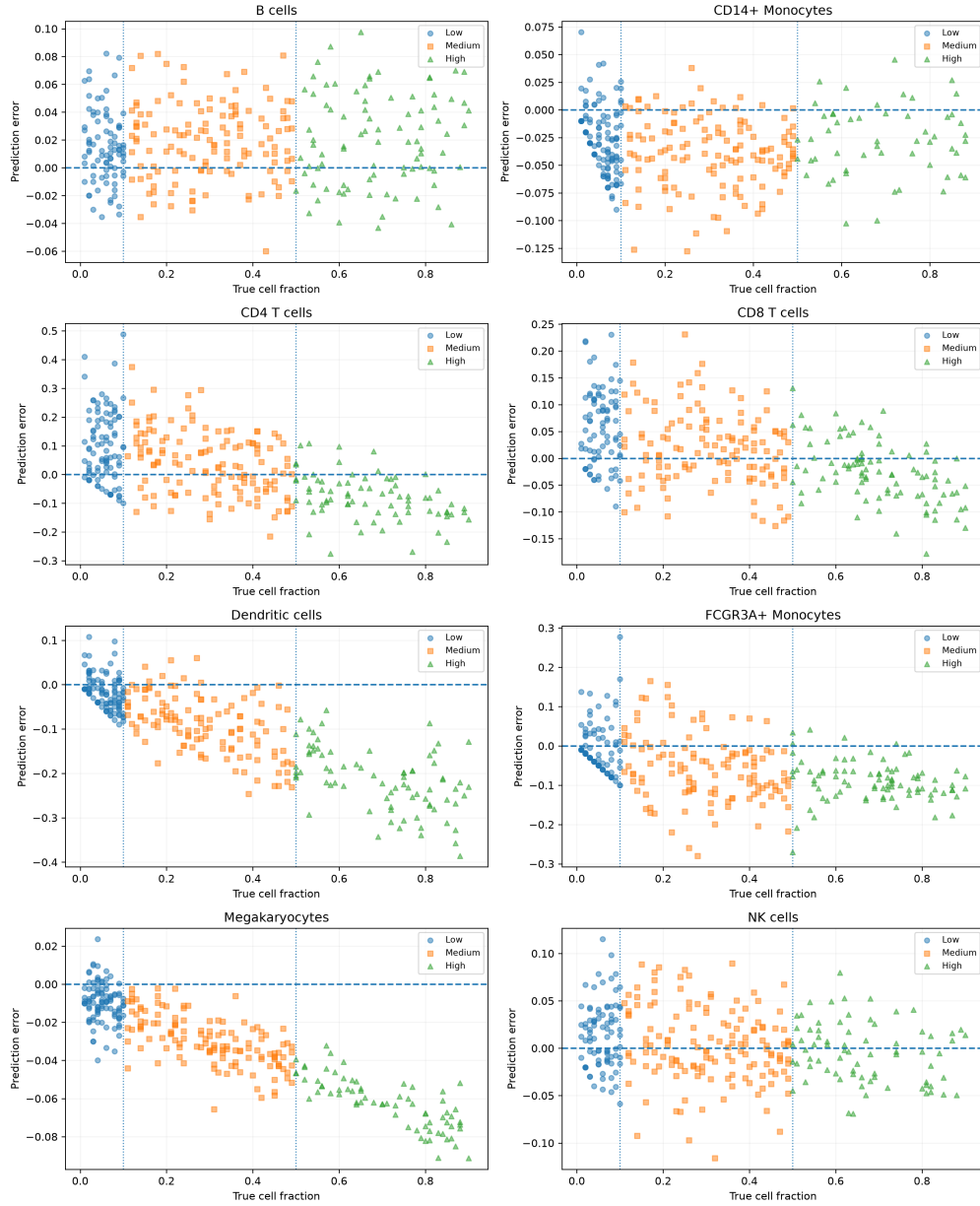


Figure 5: Prediction residuals (predicted minus true fraction) for the best experimental configuration, as a function of true cell fraction. Points are colored by abundance regime. Vertical dashed lines indicate regime thresholds.

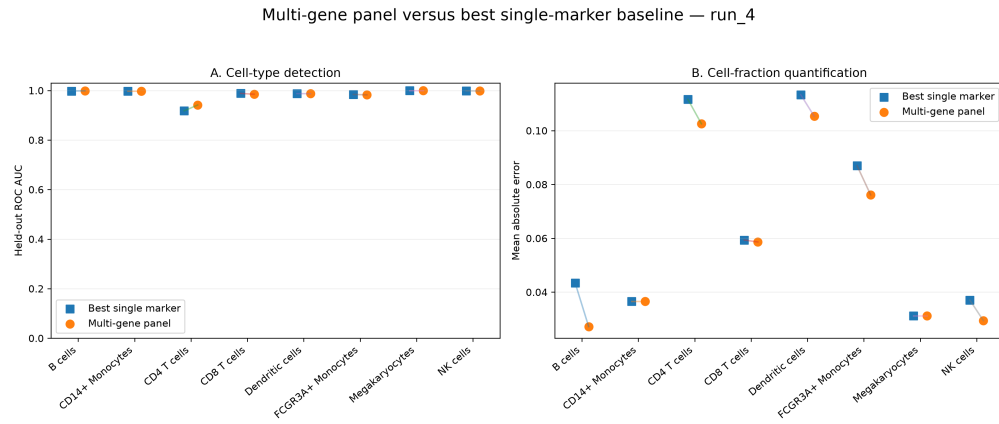


Figure 6: Comparison of multi-gene panel and best single-marker baseline for each cell type. Left: ROC AUC; Right: MAE. Each line connects the results for a given cell type.